

A Discrete Spectral Precursor from $a_1 = 5$

Exact Results from the Fibonacci Bootstrap, the 600-Cell, and the McKay Correspondence

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[Reproducible code \(GitHub\)](#)

Exact-core draft

Abstract

We isolate the exact discrete core of a larger $a_1 = 5$ framework and remove all claims that require an unproved continuum dictionary. Starting from the Fibonacci fusion rule $\tau \otimes \tau = \mathbf{1} \oplus \tau$, we prove that the bootstrap condition selecting equality between the algebraic parameter and the $SU(2)$ quantum dimension has the unique positive-integer solution $a_1 = 5$. From this integer one obtains the golden-ratio field $\mathbb{Q}(\sqrt{5})$; within the regular-convex H_4 realization class this leads to the 600-cell graph and the affine- E_8 McKay shadow of the binary icosahedral group $2I$.

We then collect the exact finite results that follow on the 600-cell and its simplicial complex: the 9-point adjacency spectrum in $\mathbb{Z}[\phi]$, the exact McKay realization of the E_8 shadow, the unique anisotropic wave operator $\square = b_1 A_{\text{fiber}} - A$ with $b_1 = 6$, the exact kernel theorem for $\square$, the exact edge-kernel decomposition $\ker(\square_1) = \rho_0 \oplus 2\rho_5$, a theorem identifying the unique compact Lie-algebra candidate compatible with the resulting 12-dimensional adjoint decomposition, the generation-count theorem on the chiral unit line of $\mathbb{Z}[\phi]$, the constructive uniqueness of the associated (a, b) assignment, the exact discrete scalar-response theorem, and the exact simplicial spectral-action coefficients $(c_0, c_1, c_2) = (2640, 14880, 55920)$.

The purpose of this paper is deliberately limited. We do not claim a full derivation of the continuum Standard Model, of 4D gravity, or of cosmological observables. Rather, we exhibit a rigid discrete spectral precursor with a nontrivial package of exact theorems, finite computations, and arithmetic uniqueness statements that any continuum interpretation would have to inherit.

Contents

1 Scope and Standards

This paper keeps only what is exact in the following sense:

1. a theorem proved from the stated assumptions;
2. an exact finite computation on the 600-cell or its simplicial complex;
3. an exact representation-theoretic or arithmetic consequence of the preceding data.

We intentionally exclude:

1. continuum identifications that still require a physical dictionary;
2. phenomenological formulas whose interpretation is not yet forced;
3. cosmological and dark-sector extrapolations.

The irreducible starting assumption is the Fibonacci fusion rule

$$\tau \otimes \tau = \mathbf{1} \oplus \tau. \quad (1)$$

Everything else in this paper is either proved from this seed together with standard mathematics, or computed exactly on a finite complex.

Theorem 1 (Rank-2 uniqueness of the non-pointed seed [?]). *Among fusion categories of rank 2 over an algebraically closed field of characteristic 0, there are only two possibilities up to monoidal equivalence: the pointed category with fusion rule $X \otimes X \cong \mathbf{1}$, and the Fibonacci category with fusion rule*

$$X \otimes X \cong \mathbf{1} \oplus X.$$

Remark 1. Theorem ?? does not derive the Fibonacci seed from physics, so the seed remains an axiom in this paper. However, it sharply limits the arbitrariness of that axiom: once one restricts to rank-2 non-pointed fusion data, the Fibonacci rule is the unique possibility.

2 Bootstrap Selection of $a_1 = 5$

Let

$$\phi(a_1) = \frac{1 + \sqrt{a_1}}{2} \quad (2)$$

and let

$$d_1(a_1) = \frac{\sin(3\pi/a_1)}{\sin(\pi/a_1)} = 1 + 2 \cos\left(\frac{2\pi}{a_1}\right) \quad (3)$$

be the $j = 1$ quantum dimension of $SU(2)$ Chern–Simons theory at level $k = a_1 - 2$.

Theorem 2 (Bootstrap vacuum selection). *The only positive integer a_1 satisfying*

$$d_1(a_1) = \phi(a_1) \quad (4)$$

is $a_1 = 5$.

Proof. Write

$$f(a_1) = \cos(2\pi/a_1) - (\sqrt{a_1} - 1)/4.$$

For $a_1 \geq 26$, $(\sqrt{a_1} - 1)/4 > 1 \geq |\cos(\cdot)|$, so $f(a_1) < 0$. A finite check on $a_1 = 1, \dots, 25$ shows $f(a_1) = 0$ only at $a_1 = 5$. Exactness at $a_1 = 5$ follows because $\cos(2\pi/5)$ is the positive root of $4x^2 + 2x - 1 = 0$. $\square$

Corollary 1 (Double embedding at $a_1 = 5$). *At the same unique value $a_1 = 5$, one also has*

$$d_{1/2}(a_1) = 2 \cos(\pi/a_1) = \phi(a_1),$$

so the nontrivial quantum dimensions are exactly ϕ .

Remark 2. The theorem is exact. What it does *not* by itself prove is that nature must realize this bootstrap physically. In this paper we use it only as the exact arithmetic entry point to the discrete structure that follows.

3 The 600-Cell, $2I$, and the E_8 Shadow

From now on set

$$a_1 = 5, \quad b_1 = a_1 + 1 = 6, \quad \phi = \frac{1 + \sqrt{5}}{2}.$$

The order of the binary icosahedral group is $|2I| = 120 = a_1!$, matching the number of vertices of the 600-cell.

Proposition 1 (Golden-ratio realization class among regular convex 4-polytopes). *Among the six regular convex 4-polytopes, the only ones whose intrinsic Coxeter/Schläfli data involve the field $\mathbb{Q}(\sqrt{5})$ are the 120-cell and 600-cell, i.e. the H_4 dual pair*

$$\{5, 3, 3\} \quad \text{and} \quad \{3, 3, 5\}.$$

Proof. The six regular convex 4-polytopes split by Coxeter type as follows:

$$\{3, 3, 3\} \leftrightarrow A_4, \quad \{4, 3, 3\}, \{3, 3, 4\} \leftrightarrow B_4, \quad \{3, 4, 3\} \leftrightarrow F_4, \quad \{5, 3, 3\}, \{3, 3, 5\} \leftrightarrow H_4.$$

The exceptional pair $\{5, 3, 3\}, \{3, 3, 5\}$ is the only one involving the parameter 5 in its Schläfli data, hence the only one whose dihedral/cosine data contain

$$\cos(\pi/5) = \frac{\phi}{2} \in \mathbb{Q}(\sqrt{5}).$$

Equivalently, the H_4 Coxeter system is the unique non-crystallographic 4D regular-polytope class whose intrinsic geometry requires the golden-ratio field. The remaining four regular convex 4-polytopes belong to the crystallographic classes A_4 , B_4 , and F_4 , and do not require $\mathbb{Q}(\sqrt{5})$ in their intrinsic Coxeter data. $\square$

Remark 3. Proposition ?? still does not prove that the bootstrap *must* be realized by a regular convex 4-polytope. What it proves is the following conditional statement: if one seeks a regular convex 4-polytope realization carrying the golden-ratio field selected by the bootstrap, then one is forced into the H_4 dual pair.

Proposition 2 (Intrinsic selection within the H_4 dual pair). *Among the six regular convex 4-polytopes, the only two lying in the golden-ratio / H_4 class are the dual pair*

$$120\text{-cell } \{5, 3, 3\} \quad \text{and} \quad 600\text{-cell } \{3, 3, 5\}.$$

Among this pair, the 600-cell is uniquely characterized by the simultaneous conditions

$$|V| = |2I| = 120, \quad \deg(v) = 12 \text{ for every vertex.} \quad (5)$$

Proof. The regular convex 4-polytopes are the 5-cell, 8-cell, 16-cell, 24-cell, 120-cell, and 600-cell. Only the dual pair (120-cell, 600-cell) belongs to the H_4 / golden-ratio class. Their basic combinatorial data are:

$$120\text{-cell} : |V| = 600, \deg(v) = 4, \quad 600\text{-cell} : |V| = 120, \deg(v) = 12.$$

Hence within the H_4 pair, the conditions (??) select the 600-cell uniquely. $\square$

Remark 4. Proposition ?? is deliberately weaker than the claim that the 600-cell is the unique regular 4-polytope “reproducing the Standard Model”. It only states an exact discrete selection result inside the intrinsic H_4 class, using vertex cardinality and local degree.

Remark 5 (Why the 120-cell does not survive the local selector). The 120-cell is not excluded from the golden-ratio background: it belongs to the same H_4 dual pair and therefore shares the $\mathbb{Q}(\sqrt{5})$ arithmetic class. What it does *not* share is the local combinatorial package used in the exact-core construction. Its vertex degree is 4 rather than 12, so it cannot supply the same 12-dimensional local candidate sector that later appears in the edge-space decomposition and in the Lie-algebra candidate theorem. Thus the 120-cell is background-compatible but local-sector incompatible with the exact-core route developed here.

3.1 Exact Spectral Data

Proposition 3 (600-cell adjacency spectrum). *The adjacency spectrum of the 600-cell graph has exactly 9 distinct eigenvalues, all in $\mathbb{Z}[\phi]$, with multiplicities*

$$\{1, 4, 9, 16, 25, 36, 9, 16, 4\}. \quad (6)$$

Equivalently, the scalar Laplacian has 9 distinct eigenvalues in $\mathbb{Z}[\phi]$.

Remark 6. This is an exact finite computation. The appearance of 9 distinct eigenvalues and square multiplicities is part of the rigid spectral package underlying the later theorems.

3.2 McKay Shadow

Proposition 4 (McKay correspondence for the binary icosahedral group). *The McKay graph of the 2-dimensional defining representation of $2I$ is the affine Dynkin diagram $\tilde{E}_8$. The 9 irreducible representations of $2I$ have dimensions*

$$1, 2, 3, 4, 5, 6, 3, 4, 2. \quad (7)$$

Remark 7. This is standard representation theory for finite subgroups of $SU(2)$; see McKay and Kostant [?, ?]. In the present paper the McKay graph is used only as exact discrete data, not as a stand-in for continuum unification.

4 The Exact Wave Operator

Fix a Hopf fibration of the 600-cell vertex set into 12 decagons arising from the left-coset decomposition of an order-10 subgroup $C_{10} \leq 2I$. Let A be the 600-cell adjacency matrix and let A_{fiber} be the adjacency matrix of the corresponding Hopf-fiber edges. Define

$$\square(c) = c A_{\text{fiber}} - A. \quad (8)$$

Theorem 3 (Exact coefficient selection). *Among integer choices of c , the nontrivial kernel condition selects uniquely*

$$\square = b_1 A_{\text{fiber}} - A = 6 A_{\text{fiber}} - A. \quad (9)$$

Remark 8. This theorem is purely discrete: it fixes the operator on the finite graph. It does *not* by itself imply a full Lorentzian continuum interpretation.

Proposition 5 (Fibration-stable vertex invariants). *Among Hopf fibrations obtained from left cosets of order-10 subgroups of $2I$, there are exactly six distinct vertex partitions of the 600-cell. For each of these six fibrations:*

1. *the unique nontrivial integer selected by the kernel condition is still $c = b_1 = 6$;*
2. *the resulting operator $\square = 6 A_{\text{fiber}} - A$ satisfies*

$$\dim \ker(\square) = 9;$$

3. *the kernel coincides with the same 9-dimensional adjacency-spectral sector,*

$$\ker(\square) = E_A(12) \oplus E_A(6\phi) \oplus E_A(6\phi'),$$

equivalently with the $2I$ -module $\rho_0 \oplus \rho_1 \oplus \rho_8$;

4. *the spectral-gap ratio satisfies*

$$\frac{\lambda_1(L_{\text{cross}})}{\lambda_1(L_{\text{fiber}})} = a_1 = 5.$$

Proof. This is an exact finite computation on the 600-cell. One enumerates all order-10 elements of $2I$, constructs the associated left-coset fibrations, deduplicates the resulting vertex partitions, and finds exactly six distinct fibrations. For each of them one computes the family $\square(c) = c A_{\text{fiber}} - A$ over integer c , obtaining the same unique nontrivial kernel value $c = 6$, the same kernel dimension 9 for $\square(6)$, the same containment in the fixed adjacency-spectral subspace

$$E_A(12) \oplus E_A(6\phi) \oplus E_A(6\phi'),$$

and the same spectral-gap ratio $\lambda_1(L_{\text{cross}})/\lambda_1(L_{\text{fiber}}) = 5$. Since this target subspace has dimension $1 + 4 + 4 = 9$, the kernel equals it for every one of the six fibrations. $\square$

Remark 9. Proposition ?? isolates the part of the construction that is genuinely fibration-stable. It does *not* claim that every finer invariant is independent of the fibration: for example, the positive/negative signature split of $\square$ varies across the six fibrations, while the kernel sector and gap ratio do not.

Theorem 4 (Galois kernel theorem). *For any Hopf fibration of the above left-coset type and $\square = 6A_{\text{fiber}} - A$,*

$$\ker(\square) = \rho_0 \oplus \rho_1 \oplus \rho_8, \quad (10)$$

so $\dim \ker(\square) = 9$.

Remark 10. Theorem ?? is now uniform across all six distinct left-coset Hopf fibrations of the 600-cell. What remains fibration-dependent is not the kernel sector itself but finer signature data of $\square$.

5 Exact Edge-Space Structure

Let $L(G)$ be the line graph of the 600-cell. Define on edge space the exact analogue

$$\square_1 = L_{\text{cross}} - a_1 L_{\text{fiber}}. \quad (11)$$

Theorem 5 (Edge kernel decomposition). *The kernel of $\square_1$ has dimension 13 and decomposes under $2I$ as*

$$\ker(\square_1) = \rho_0 \oplus 2\rho_5. \quad (12)$$

Proposition 6 (Factorization through A_5). *The action of $2I$ on the 12 Hopf fibers factors through the quotient*

$$A_5 \cong 2I/\{\pm 1\}. \quad (13)$$

Proof. The central element $-1 \in 2I$ acts by the antipodal map on vertices and preserves each Hopf fiber setwise, hence acts trivially on the fiber labels. Therefore the permutation action on the 12 fibers factors through the quotient by $\{\pm 1\}$. $\square$

Corollary 2 (Exact 12-dimensional fiber permutation module). *The permutation representation of A_5 on the 12 Hopf fibers decomposes exactly as*

$$12 = \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}. \quad (14)$$

Remark 11. The preceding corollary is an exact finite statement about the fiber permutation module. It does not by itself produce a local gauge connection or a continuum Lie bracket.

Theorem 6 (Unique compact Lie-algebra candidate). *Let G be a compact connected Lie group of dimension 12 whose adjoint representation, restricted to A_5 , decomposes as*

$$\text{ad}(G)|_{A_5} = \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}. \quad (15)$$

Then

$$\mathfrak{g} \cong \mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3). \quad (16)$$

Proof. The trivial summand forces an abelian $\mathfrak{u}(1)$ factor, leaving a compact semisimple Lie algebra of dimension 11. The only simple compact Lie algebras with dimension at most 11 are $\mathfrak{su}(2)$ (dimension 3), $\mathfrak{su}(3)$ (dimension 8), and $\mathfrak{sp}(2)$ (dimension 10). The only decomposition of 11 into allowed simple dimensions is $3 + 8$, hence $\mathfrak{su}(2) \oplus \mathfrak{su}(3)$. $\square$

Remark 12. The theorem identifies a unique compact Lie-algebra *candidate* compatible with the exact discrete 12-dimensional decomposition. If one further fixes the simply connected semisimple part, the corresponding compact group is $U(1) \times SU(2) \times SU(3)$ up to finite central quotient. This paper does not claim that the full nonabelian continuum gauge theory has thereby been derived.

6 Exact Generation Count on the Chiral Unit Line

Theorem 7 (Generation-count theorem on the chiral unit line). *On the line $a = 1$ of $\mathbb{Z}[\phi]$, exactly three values of b give unit elements:*

$$1 + b\phi \in \mathbb{Z}[\phi]^\times \iff b \in \{0, 1, 2\}. \quad (17)$$

Proof. By Dirichlet's unit theorem for the real quadratic field $\mathbb{Q}(\sqrt{5})$, all units are $\pm\phi^k$. Since

$$\phi^k = F(k-1) + F(k)\phi,$$

the condition $1 + b\phi = \phi^k$ requires $F(k-1) = 1$. The Fibonacci sequence takes the value 1 exactly at indices $-1, 1, 2$, hence $k \in \{0, 2, 3\}$ and $b \in \{0, 1, 2\}$. $\square$

Remark 13. This theorem is exact arithmetic. It yields three stable chiral unit sectors. Whether these should be interpreted as the three physical fermion families is a separate continuum-flavor dictionary and is not needed here.

7 Constructive (a, b) Uniqueness

For any mass exponent n , solutions of $n = 5a + 6b$ are parametrized by integers. The exact quadratic norm on $\mathbb{Z}[\phi]$ is

$$N(a + b\phi) = a^2 + ab - b^2. \quad (18)$$

Theorem 8 (Constructive uniqueness of the exact assignment). *For the exponent set*

$$\{0, 3, 5, 11, 11, 16, 17, 19, 26\}, \quad (19)$$

the assignment

$$(a, b) = (0, 0), (3, -2), (1, 0), (1, 1), (1, 1), (2, 1), (1, 2), (4, 1), (-1, 4) \quad (20)$$

is uniquely obtained by the following exact constructive chain:

1. *the McKay main-chain weights are fixed;*
2. *each edge jump Δn is lifted by the unique minimizer of $|\Delta a| + |\Delta b|$ under $5\Delta a + 6\Delta b = \Delta n$;*
3. *the root is fixed at $(0, 0)$ and the lifts are integrated along the graph;*
4. *the heavy prime sector is closed by the exact Galois relation*

$$z_b = \phi \sigma(z_t). \quad (21)$$

Remark 14. This is an exact theorem about the arithmetic and graph structure. In the present paper we make no stronger claim than that the assignment is uniquely fixed by the discrete precursor once the exponent set is given.

8 Exact Discrete Scalar Response

Let $d_0 : \mathbb{R}^V \rightarrow \mathbb{R}^E$ and $d_1 : \mathbb{R}^E \rightarrow \mathbb{R}^F$ be the simplicial coboundary operators of the 600-cell complex. Let

$$\Delta_0 = d_0^T d_0, \quad B = d_0 d_0^T. \quad (22)$$

Theorem 9 (Exact discrete scalar response). *The Moore–Penrose pseudoinverses satisfy*

$$B^+ d_0 = d_0 \Delta_0^+. \quad (23)$$

For a point source at a vertex v_0 , define

$$\Phi = \Delta_0^+ \delta_{v_0}, \quad h = B^+ d_0 \delta_{v_0}. \quad (24)$$

Then

$$h = d_0 \Phi, \quad d_1 h = 0, \quad d_0^+ h = \Phi. \quad (25)$$

Proof. For any matrix A , one has $A^+ = (A^T A)^+ A^T = A^T (A A^T)^+$. Taking $A = d_0$ yields the identity. The three conclusions follow immediately. $\square$

Remark 15. The theorem proves an exact discrete scalar-response identity. It does *not* prove the 4D continuum PPN statement $\gamma = 1$.

9 Exact Simplicial Spectral-Action Data

Let

$$D = d + d^* \quad (26)$$

be the Dirac-type operator on the full simplicial complex of the 600-cell. Then

$$\dim \mathcal{H} = 120 + 720 + 1200 + 600 = 2640.$$

Proposition 7 (Exact coefficient triple). *The first three exact spectral-action coefficients are*

$$(c_0, c_1, c_2) = (2640, 14880, 55920). \quad (27)$$

Equivalently, after dividing by $2N = 240$ one gets the reduced triple

$$(A_0, A_1, A_2) = (11, 62, 233). \quad (28)$$

Proposition 8 (Exact Diophantine identity). *The reduced coefficients satisfy*

$$2A_1^2 + 1 = 3A_0 A_2. \quad (29)$$

Proof. Direct substitution gives

$$2 \cdot 62^2 + 1 = 7689 = 3 \cdot 11 \cdot 233.$$

$\square$

Remark 16. The coefficient triple and the identity are exact finite data. Their interpretation as cosmological, Einstein–Hilbert, and Yang–Mills terms in a continuum action remains a separate structural step and is not used here.

10 What This Paper Does Not Claim

This exact-core paper does *not* claim:

1. a derivation of the physical fine-structure constant;
2. a derivation of the full continuum Standard Model Lagrangian;
3. a derivation of 4D general relativity;
4. a derivation of neutrino, dark-sector, or cosmological observables.

What it does claim is that, once the Fibonacci bootstrap is adopted and the regular-convex H_4 realization class is fixed, there exists a rigid package of exact discrete results centered on $a_1 = 5$, the 600-cell, the binary icosahedral group, the affine- E_8 McKay shadow, a uniquely selected anisotropic wave operator, an exact 12-dimensional edge sector with a unique compact Lie-algebra candidate, a three-slot chiral unit structure on $\mathbb{Z}[\phi]$, an exact constructive arithmetic assignment, an exact scalar-response theorem, and an exact simplicial spectral-action coefficient triple.

11 Upgrade Criteria for Future Versions

Any statement considered for inclusion in a future exact-core version should satisfy the following checklist:

1. **Exact discrete input:** the underlying object must be a theorem or an exact finite computation.
2. **Physical target:** the candidate continuum object being represented must be stated explicitly.
3. **Mechanism:** there must be a mathematically controlled map, reduction, or standard physical mechanism connecting the discrete object to the claimed continuum target.
4. **Scope:** if the mechanism stops at Lie algebra, spectrum, or response level, the paper should not claim more than that.
5. **Falsifiability:** the proposed interpretation should imply an independent observable or obstruction.

Statements failing item (1) do not enter this paper. Statements satisfying (1) but not yet (2)–(5) belong in a separate structural or speculative document rather than in the exact core.

12 Conclusion

The exact core of the $a_1 = 5$ framework is already substantial without any aggressive continuum interpretation. It consists of:

1. a unique integer selected by a bootstrap theorem;

2. a conditionally selected finite geometric realization on the 600-cell inside the regular-convex H_4 class;
3. a rigid representation-theoretic shadow governed by affine E_8 ;
4. an exact anisotropic wave operator with nontrivial vertex and edge kernels, together with a fibration-stable vertex-level kernel dimension and gap ratio;
5. a unique compact Lie-algebra candidate selected by the exact 12-dimensional decomposition;
6. an exact three-slot chiral unit structure on $\mathbb{Z}[\phi]$;
7. exact constructive arithmetic assignments in the flavor lattice;
8. an exact discrete scalar-response theorem;
9. exact simplicial spectral-action coefficients and identities.

These results define a nontrivial discrete spectral precursor. Whether and how this precursor should be completed into continuum particle physics remains open, but any such completion would have to respect the exact structure recorded here.

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